

Christos Koutras

Postdoctoral Research Associate · NYU VIDA

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Postdoctoral researcher at NYU VIDA building data integration solutions for practitioners. Work spans data matching at the schema and value levels, metadata generation, and the composite harmonization workflows that bring these together.

Research Interests Data Integration · Schema & Data Matching · Metadata Generation · AI for Data Harmonization

Languages Greek (native) · English (C2 CPE, U. Michigan) · German (B1, Goethe Institut)

EDUCATION

PhD, Computer Science

2018 – 2024

Delft University of Technology (TU Delft), Netherlands

Research: *Tabular Schema Matching for Modern Settings*.

Supervisors: Asterios Katsifodimos, Christoph Lofi, Geert-Jan Houben.

MPhil, Computer Science

2016 – 2018

Hong Kong University of Science & Technology (HKUST), Hong Kong

Thesis: *Query Estimation Structures in Geo-Social Networks*.

Supervisor: Dimitris Papadias. GPA: 3.94 / 4.

Diploma (M.Eng), Electrical & Computer Engineering

2009 – 2015

National Technical University of Athens (NTUA), Greece

Major: Computer Science. GPA: 8.24 / 10.

Thesis: *Monitoring Continuous k-Nearest Neighbor Queries Over Moving Uncertain Objects*.

Supervisors: Kostas Patroumpas, Yannis Vasileiou.

EXPERIENCE

Postdoctoral Research Associate

Nov 2024 – Present

New York University · VIDA Research Center, Brooklyn, NY

Visualization and Data Analytics Research Center, Tandon School of Engineering.

Applied Scientist Intern

Dec 2023 – Mar 2024

Amazon Web Services, SF Bay Area, USA

Optimized LLM-usage in data wrangling pipelines.

Applied Scientist Intern

Apr – Jul 2022

Amazon Web Services · GraphML team, SF Bay Area, USA

Self-supervised join discovery among tabular data.

Visiting Student

Jul – Aug 2017

Oxford University, United Kingdom

Query optimization with access patterns under integrity constraints, with George Konstantinidis.

TEACHING

Course Section Leader · DS-GA 3001 Data Engineering

Jan – Apr 2026

New York University, Manhattan, NY

Organizing and delivering the course's lab sessions.

Course Coordinator · CSE1500 Web & Database Technology

Nov – Dec, 2018 – 2023

Delft University of Technology, Netherlands

Organized DB lab sessions, taught tutorials, designed database assignments.

Course Coordinator · CSE1505 Information & Data Management

Feb – Apr, 2018 – 2023

Delft University of Technology, Netherlands

Organized labs and tutorials; designed and graded lab assignments and exam tasks.

Teaching Assistant · COMP 3311 Database Management Systems

Feb – May 2018

Hong Kong University of Science & Technology, Hong Kong

Tutorials on functional dependencies, normalization, memory hierarchy, indexing & hashing, join algorithms, query processing and optimization.

Designed and graded assignments on greedy algorithms, dynamic programming, minimum spanning trees, shortest path algorithms.

Tutorials on ER diagrams, relational data model, relational algebra, SQL, functional dependencies and normalization.

PROJECTS

Valentine

ICDE 2021; VLDB Demo 2021

Open-source library and benchmark for schema matching on tabular data. Core maintainer of the `valentine` Python package.

OmniMatch

VLDB 2025

Joinability discovery in data products via graph neural networks; developed during an AWS internship.

BDI-Kit

Patterns; SIGMOD Demo 2026

Toolkit pairing classical matchers with LLM-driven workflows for biomedical data harmonization.

BDIViz

VIS 2025; SIGMOD Demo 2026

Interactive visualization system for schema matching verification with LLM-powered validation.

PUBLICATIONS

Aligning Real-World Document Schemas to Fine-Grained Concept Vocabularies

CIKM 2026

Enver Menadjeiev, Isaac Fisher, Kai Spicer, **Christos Koutras**, Andrew Bell

Interoperability in Healthcare: A Primer and New Frontiers

VLDB 2026

Andra Ionescu, Paris Carbone, Sebastiaan Meijer, **Christos Koutras**, Juliana Freire, Katja Hose, Jayanth Raghothama

BDIViz in Action: Interactive Curation and Benchmarking for Schema Matching Methods

SIGMOD 2026

Eden Wu, **Christos Koutras**, Juliana Freire.

BDI-Kit Demo: A Toolkit for Programmable and Conversational Data Harmonization

SIGMOD 2026

Roque Lopez, Yurong Liu, **Christos Koutras**, Juliana Freire.

BDI-Kit: An AI-powered Toolkit for Biomedical Data Harmonization

Patterns

Roque Lopez, Aecio Santos, **Christos Koutras**, Juliana Freire.

OmniMatch: Joinability Discovery in Data Products

VLDB 2025

Christos Koutras, Jiani Zhang, Xiao Qin, Chuan Lei, Vasileios Ioannidis, Christos Faloutsos, George Karypis, Asterios Katsifodimos.

BDIViz: An Interactive Visualization System for Biomedical Schema Matching with LLM-Powered Validation

VIS 2025

Eden Wu, Dishita G Turakhia, Guande Wu, **Christos Koutras**, Sarah Keegan, Wenke Liu, Beata Szeitz, David Fenyo, Claudio Silva, Juliana Freire.

Large Language Models for Data Discovery and Integration: Challenges and Opportunities

IEEE Data Engineering

Bulletin

Juliana Freire, Grace Fan, Benjamin Feuer, **Christos Koutras**, Yurong Liu, Eduardo Pena, Aecio Santos, Claudio Silva, Eden Wu.

SiMa: Effective and Efficient Matching Across Data Silos Using Graph Neural Networks

TRL @ NeurIPS 2022

Christos Koutras, Rihan Hai, Kyriakos Psarakis, Marios Fragkoulis, Asterios Katsifodimos.

Data Lakes: A Survey of Functions and Systems

IEEE TKDE

Rihan Hai, **Christos Koutras**, Christoph Quix, Matthias Jarke.

Amalur: Data Integration Meets Machine Learning

ICDE 2023

Rihan Hai, **Christos Koutras**, Andra Ionescu, Ziyu Li, Wenbo Sun, Jessie van Schijndel, Yan Kang, Asterios Katsifodimos.

Amalur: Next-generation Data Integration in Data Lakes

CIDR 2022

Rihan Hai, **Christos Koutras**, Andra Ionescu, Marios Fragkoulis, Asterios Katsifodimos.

Valentine in Action: Matching Tabular Data at Scale

VLDB 2021

Christos Koutras, Kyriakos Psarakis, George Siachamis, Andra Ionescu, Marios Fragkoulis, Angela Bonifati, Asterios Katsifodimos.

Valentine: Evaluating Matching Techniques for Dataset Discovery

ICDE 2021

Christos Koutras, George Siachamis, Andra Ionescu, Kyriakos Psarakis, Jerry Brons, Marios Fragkoulis, Christoph Lofi, Angela Bonifati, Asterios Katsifodimos.

REMA: Graph Embeddings-based Relational Schema Matching

SeaData Workshop @ EDBT/ICDT 2020

Christos Koutras, Marios Fragkoulis, Asterios Katsifodimos, Christoph Lofi.

Data as a Language: A Novel Approach to Data Integration

VLDB PhD Workshop 2019

Christos Koutras.

Probabilistic k-Nearest Neighbor Monitoring of Moving Gaussians

SSDBM 2017

Kostas Patrourmpas, **Christos Koutras**.

AWARDS & CERTIFICATIONS

IEEE ICDE Distinguished PC Award

2026

Awarded to PC Committee members for outstanding contributions in reviewing and discussions.

University Teaching Qualification (UTQ)

2023

Certifies the didactic competences of lecturers in academic education across the Netherlands. Recognized by all fourteen Dutch universities.

PROFESSIONAL SERVICE

PC Member DBML 2022–2024 / 2026 (workshop with ICDE), ICDE 2024 Demonstration Track, ECML/PKDD 2024 Applied Data Science Track, VLDB 2026/2027, ICDE 2026/2027, SIGMOD 2026, AIDB 2026 (VLDB Workshop), NOVAS 2026 (VLDB Workshop).

Reviewer for Journals Data & Knowledge Engineering (DKE), Patterns (Cell Press), Nature Communications, Information Systems.

External Reviewer VLDB 2023/2024, IEEE ICDE 2019/2023, EDBT 2024, ACM SIGMOD 2018/2021, ACM SIGSPATIAL 2017, SSTD 2017.